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ORIGINAL ARTICLE

Using sugar cane vinasse as a biofertilizer: effects on soil microbial community due to periodical application

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Abstract

Bioethanol production from sugarcane generates by-products such as vinasse, with a high content of organic matter and potassium. While vinasse can be used as a fertilizer, its effect on the soil and its microbiota is unknown. In this work, we evaluated the effect of systematic vinasse application on microbial communities of soils used for sugarcane cultivation. The assays were carried out in the field and controlled conditions using the same soil type previously irrigated only with water or with vinasse. In the field, soil samples were taken prior to summer irrigation and one month later during years. No significant differences were found in the abundance of most microbial populations nor in the microbial activity by the vinasse treatment. However, in some sampling dates, populations of phosphate solubilizing bacteria, *Bacillus* spp. and ammonifying bacteria were significantly lower in soil with systematic vinasse irrigation compared to the control. Under controlled conditions, the treatments included two doses of vinasse and a control without vinasse. The abundance of heterotrophic bacteria and fungi increased in both soils with vinasse addition compared to the control, and at 180 days an increase in ammonifying bacteria was observed. Although microbial populations remained higher with the vinasse treatments, their activity decreased over time, with more evident effects on soils without application history. According to our results, vinasse application could have temporary effects on microbial communities, depending on soil type and doses, so local research is needed to create technical standards that regulate the use of vinasse on agricultural soils.

Keywords Fertirrigation · Industrial waste · Soil improvement · Soil quality · Biofuels

Introduction

During the last decades the energy system has been moving towards to the use of biofuels, not only by the demand for energy worldwide but also due to concerns about climate change. The production of ethanol from sugarcane is increasingly becoming an alternative source of energy worldwide and is considered both economically and environmentally sustainable depending on the ability to recycle the waste products that arise from the process (Braga do Carmo et al. 2012). These wastes may contain large amounts of organic carbon and a wide range of nutrients, which makes them attractive for its use as soil amendments or plant fertilizers (Cayuela et al. 2010; Galvez et al. 2012). However, vinasse, the major waste product of ethanol production can be a significant source of emission of greenhouse gases (Cipriano et al. 2019). These emissions can result from aerobic or anaerobic decomposition of vinasse´s organic matter and can occur during transport, storage or after its application in soils (Gonçalves de Oliveira et al. 2013).

Vinasse is a dark brown liquid effluent characterized by having a high content of organic matter, potassium, calcium, magnesium, sulfur and nitrogen; and its pH ranges between 3.5 and 5 (Montiel-Rosales et al. 2022; Cerri et al. 2020; Rulli et al. 2020; Christofoletti et al. 2013; España-Gamboa et al. 2012). Due to its composition in nutrients beneficial for plant growth, vinasse is used in different countries as a plant fertilizer, thus preventing it to become an environmental pollutant (Montiel-Rosales et al. 2022).

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Due to the large increase in agricultural productivity and agroindustrial practices employed in recent decades, land degradation has reduced 40% the productive capacity of soils, so applying to the soil compounds-rich organic matter has become a promising agricultural practice to replenish organic matter and improve the structure and fertility of soils (Bustamante et al. 2010; Girvan et al. 2003). The irrigation with vinasse is a fairly common practice in crop areas of sugarcane in Brazil, where satisfactory results have been obtained in relation to organic matter, pH and nutrients for crops (Bebé et al. 2009).

In Uruguay, sugarcane cultivation covers an approximately area of 11,500 ha located in Bella Unión—Artigas, where Alcoholes del Uruguay S.A. (ALUR) Sugarmill is located. Bioethanol production is carried out there and 6–7 L of vinasse are generated per liter of bioethanol produced, with a productive capacity of 120 m^3/day of ethanol. Due to this, large volumes of stillage are generated so that must be managed properly (Takata 2020). Soil vinasse application has started to be used by ALUR (Alcoholes del Uruguay S.A.) in 2012, an activity currently authorized by DINAMA (Dirección Nacional de Medio Ambiente) as a way to recycle this waste, yet conditioned to studies and monitoring of its effects on the environment. So far, in our country, there are no technical standards in which criteria and procedures are specified for the vinasse application in agricultural soils as it occurs in other countries such as Brazil (CETESB 2015).

In most of published works, the use of vinasse is assessed analysing the impact on the physicochemical properties of the soil and crop productivity (Del Pino et al. 2017; Hazbavi and Sadeghi 2016; Jiang et al. 2012; Barbazán et al. 2011; Pessoa de Barros et al. 2010; Parnaudeau et al. 2008). However, its effect on soil microorganisms, responsible for vinasse degradation, is often overlooked. These microorganisms play a fundamental role in biogeochemical cycles, contribute to plant nutrition and health, soil structure and fertility, and are essential for the productivity and stability of the ecosystem (Kennedy 1999). Furthermore, microbial diversity may have an important role in soil quality maintenance. Microbial communities can respond after different types of disturbances, hence the characterization of its composition and/or structure could be used as soil quality indicators (Lambias et al. 2005).

Vinasse applications may affect soil microorganism community and the consequences of its application depends on factors related to soil and crop type, type of management and vinasse characteristics, so local research on these aspects are needed.

The objective of this work was to evaluate the effect of the periodical use of vinasse as a biofertilizer on the microbial community of the soil, in order to validate an alternative use of this residue. For this, the effect of the application of the vinasse on the abundance and microbial activity was evaluated using two different strategies: a field assay and a laboratory test under controlled conditions.

Materials and methods

Field assay

Soil sampling

Soil samples were collected from fields belonging to ALUR (Alcoholes del Uruguay S.A.) located at Bella Unión, Artigas-Uruguay (30°19′ S and 57°36′ W) (Fig. 1). The region has a humid subtropical climate, with annual average temperature of 19.6 °C, high and low average of 26.2 and 13.7 °C, respectively. Average precipitation is 1366 mm per year, the lowest in August with 66 mm and the highest in April with an average of 147 mm.

The assay was carried out in two sites (site A and site B) that have the same type of soil (L1: sandy clay loam), where a "baseline" of several microbiological variables had been determined in 2011. Microorganism quantification was carried out by plate counting or using the most probable number (MPN) method, while microbial activity was measured using respirometry technique. Physicochemical properties were also analyzed in 2011 before starting annual irrigation (Table 1).

Site A received traditional water irrigation and conventional chemical fertilization (1.3 kg of 26-0-26). Site B was irrigated annually with an equivalent vinasse concentration of 150 m^3/ha approximately (CETESB 2015) in summer periods during 6 years. Samples were taken prior to the annual vinasse application and after one month of being irrigated throughout the summer, at 4th, 5th and 6th year.

Three soil samples per site, consisting on a mix of 10 randomly collected cores (2 cm diameter × 10 cm depth), were taken before and after vinasse application. Samples were air-dried, sieved to 2 mm, and stored at 4 °C for no longer than two months for culture-dependent analysis, and at −80 °C for molecular analysis.

Culturable bacteria

For each soil sample, 5 g of soil were mixed with 45 mL of 0.1% (w/v) sodium pyrophosphate solution in 125 mL bottles and stirred for 30 min at 150 rpm and 30 °C in an orbital shaker. The suspension obtained (dilution 0) was serially diluted. Aliquots of 10 µL from three dilutions (10^{-3}, 10^{-4} and 10^{-5}) were spread by triplicate on tenfold diluted tryptic soy agar (1/10 TSA) (Smit et al. 2001) to enumerate culturable heterotrophic bacteria and aliquots of 100 µL of two dilutions (10^{-3} and 10^{-4}) were spread in duplicate on Starch Casein Agar (SCA) (Leoni and Ghini 2003) to enumerate

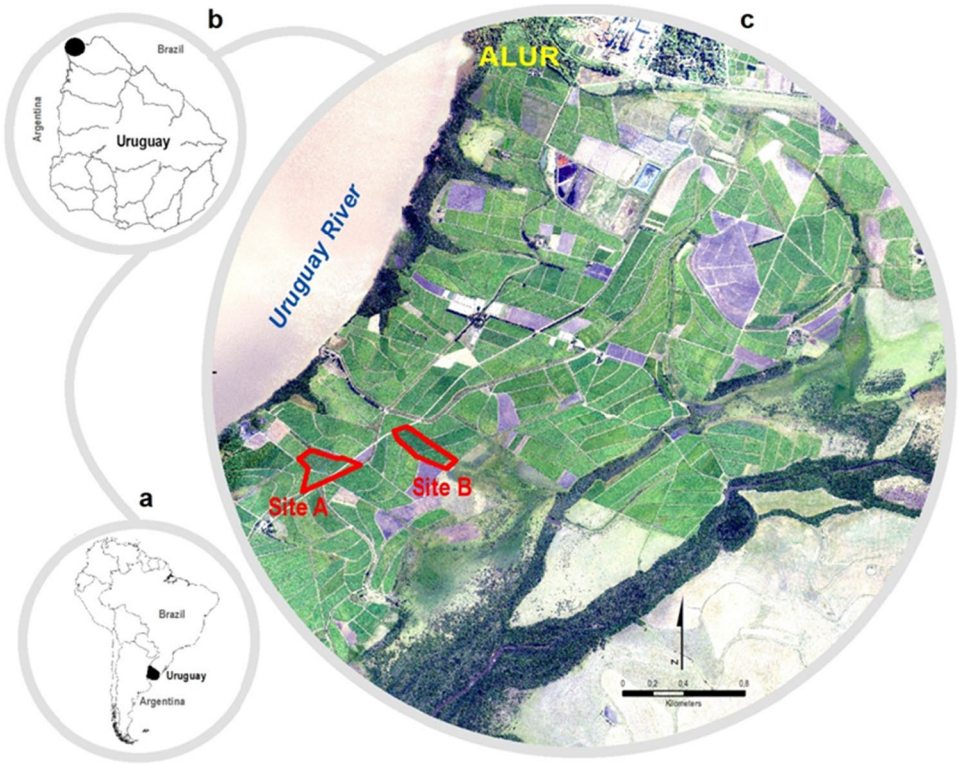

Fig. 1 **a** South America, Uruguay highlighted. **b** Location of Bella Unión—Artigas, within Uruguay. **c** Location of study sites in Bella Unión

Table 1 Soils physicochemical properties

Variable	Sand	Silt	Clay	Organic C	pH H_2O	Ca^{2+}	Mg^{2+}	K^+	Na^+	CEC[a] pH7	Base Saturation pH7
	%			g/kg	…	$cmol_c$/kg					%
Site B	56	18	26	14	4.5	12.38	2.49	0.32	0.47	21.57	72.6
Site A	43	30	27	16	5	12.09	1.98	0.38	0.39	20.74	71.5

[a]Cation-exchange capacity

actinobacteria. Plates were incubated at 25 °C for 7 days. Heterotrophic bacteria and typical actinobacterial colonies were counted after 7 days. To enumerate fluorescent *Pseudomonas* 100 µL from dilution 0 and 10^{-1} were spread in duplicate on King´s B (KB) agar (King et al. 1954) supplemented with 12.5 µg/mL of chloramphenicol and 50 µg/mL of ampicillin. Plates were incubated at 25 °C for 48 h and fluorescent colonies under ultraviolet light were counted. For spore forming bacteria (mainly *Bacillus* spp.) two dilutions (10^{-2} and 10^{-3}) were incubated at 85 °C for 30 min to remove vegetative cells (Stevenson and Segner 1992) and 100 µL of each were spread in duplicate on tenfold diluted tryptic soy agar (1/10 TSA). Plates were incubated at 25 °C for 48 h and all colonies (originating from spores) were counted. For counting free-living nitrogen fixing bacteria, aliquots of 1 mL from two dilutions (10^{-3} and 10^{-4}) were seeded by inclusion in nitrogen-fixing bacteria (Nfb) plates (Döbereiner 1980). A 2% agar-water overlay supplemented with 100 µg/mL of cycloheximide was added to plates to decrease the diffusion of oxygen in the medium. Plates were incubated at 25 °C for 7 days and all grown colonies were counted. Also, aliquots of 100 µL from 10^{-1} and 10^{-2} dilutions were spread on National Botanical Research Institute's Phosphate (NBRIP) media to enumerate phosphate solubilizing bacteria (PSB). Plates were incubated at 25 °C and the number of phosphate solubilizing bacteria (PSB) was obtained by counting colonies that formed a solubilization halo in NBRIP (National Botanical Research Institute's Phosphate) at day 7. All media contained 100 µg/mL of cycloheximide to inhibit fungal growth. To quantify ammonifying microorganisms by MPN (most probable number) technique, aliquots of 100 µL from three dilutions (10^{-3}, 10^{-4} and 10^{-5}) were inoculated by triplicate in 4% (w/v) peptone broth. Tubes were incubated during 48 h at 150 rpm and 30 °C in an orbital shaker and ammonia was detected using Nessler reactive (Seeley et al. 1991).

Filamentous fungi and yeasts abundance were quantified by spreading aliquots of 100 µL from two dilutions (10^{-1} and 10^{-2}), in duplicate, on Malt Agar 2% (w/v) plates supplemented with chloramphenicol 100 µg/mL and pH 4.5 adjusted with lactic acid (Bettucci and Roquebert 1995). Plates were incubated at 25 °C and the number of colonies of each group was determined at 72 h.

In order to express the microbial populations as colony forming units (CFU) or most probable number (MPN) per g of dry soil, the moisture content of the samples was estimated by drying at 100ºC during 48 h.

Microbial respiration

For each soil sample, 50 g of soil were watered to reach 75% of its field capacity. Each sample was placed in an airtight container along with a bohemian vessel containing 5 mL of 0.25 N NaOH. Two flasks containing only NaOH were used as control. After 48 h of incubation at 25 °C titration with 0.1 N HCl and phenolphthalein as indicator was carry out. Calculations were made to determine the CO_2 breathed by the soil microorganisms per gram of dry soil (Anderson 1982).

DNA extraction

The DNA of each sample was obtained from 0.25 g of soil using the Power Soil DNA Isolation Kit (Mo Bio), according to the manufacturer's instructions. The DNA extracted was visualized by electrophoresis in 0.8% agarose gels (w/v) with 1× TAE buffer (Tris–Acetic buffer-EDTA), at 80 V for 1 h, to check its integrity and determine its concentration.

Real time PCR

To determine total bacteria present in soil samples, the number of copies of 16S rRNA gene was quantified by Real Time PCR using the Rotor Gene SYBR green PCR commercial kit (QIAGEN) and a Rotor-Gene 6000 device (Corbett brand). Specific primers (Ba519f/Ba907r) were used to amplify the 16S rRNA gene belonging to the *Bacteria* domain under conditions reported in the literature (Lueders et al. 2004). Calibration curves were made using purified genomic DNA, extracted with a conventional DNA extraction method (Sambrook et al. 1987), from a *Pseudomonas fluorescens* strain. For these calibration curves three concentrations (1.5, 0.15 and 0.015 ng/µL) of the genomic DNA were used and each point of the curve was performed in triplicate. The reaction was carried out in a total volume of 12.5 µL using the SBYR Green PCR Master Mix 1×, containing 1 µM of each specific primer and 2.5 µL of DNA extracted from the soil with the appropriate dilution, so that its concentration fitted within the concentration range used for the calibration curve. All reactions were done in triplicate. The cycling conditions consisted of an initial denaturation of 5 min at 95 °C followed by 40 cycles of 95 °C for 10 s, a hybridization step at 52 °C for 10 s and an extension step at 72 °C for 20 s.

Controlled conditions assay

Soil samples from sites A and B were incubated under controlled laboratory conditions (25 °C and field capacity). Soil samples were taken from the first 20 cm depth and subsequently sieved to remove residual plant and inert materials. Each soil sample (1 kg) was mixed with different vinasse doses (2 treatments equivalent to 125 and 250 m^3/ha) and a control without vinasse (water at field capacity). The mixtures were placed in plastic trays (30*15*5 cm) with small holes that allowed gas exchange (aerobic conditions). Samples of 30–40 g of soil were taken at day 14, 33, 117 and 180 to study the abundance of heterotrophic bacteria, actinobacteria, filamentous fungi, and of ammonifying bacteria by the techniques described in "Culturable bacteria" section.

Statistical analysis

Data were analyzed with the Statistica 5.0 software (Statsoft). Normal distribution of data was verified with Shapiro–Wilk's test. Homogeneity of variances was assessed by Hartley's, Cochran's and Bartlett's tests. Values of bacterial counts (transformed to $\log_{10}$), most probable number (MPN), gene copy number by Real Time PCR (transformed to $\log_{10}$) and microbial activity were compared by ANOVA and the Tukey's honestly significant difference (HSD) test, only for the cases where the data distribution was normal and the variances homogeneous. Otherwise, the non-parametric Kruskal–Wallis analysis of variance and the Mann–Whitney test were performed. A significance level of $p < 0.05$ was considered.

Results

Field assay

For most microbial variables analyzed in the first spring, no significant differences between sites were found regarding microbial abundance and activity measured as respiration (Table 2). However, populations of phosphate solubilizing bacteria and *Bacillus* spp. were significantly lower in the site with annual vinasse irrigation for 4 years (site B) compared to the control site (site A). In the first autumn, heterotrophic bacteria population showed a small decrease in the site treated with vinasse respect to the control site. For the other microorganisms no significant differences were found

Table 2 Microbiological variables at sites A and B before and after summer irrigation (Year 1)

Variable	Year 1			
	Spring (before summer irrigation)		Autumn (after summer irrigation)	
	Site A (control plot)	Site B (vinasse 150 m^3/ha)	Site A (control plot)	Site B (vinasse 150 m^3/ha)
	CFU[a]/g dry soil		CFU[a]/g dry soil	
Heterotrophic bacteria	2.1×10^7 a	1.1×10^7 a	8.3×10^6 a	5.1×10^6 b
Actinobacteria	1.2×10^6 a	1.0×10^6 a	1.2×10^6 a	1.3×10^6 a
N fixing bacteria	6.5×10^5 a	8.3×10^5 a	1.3×10^6 a	1.1×10^6 a
Pseudomonas fluorescent	2.0×10^4 a	1.6×10^4 a	5.0×10^3 a	2.1×10^4 a
Phosphate solubilizing bacteria	3.0×10^5 a	2.0×10^5 b	4.5×10^4 a	1.3×10^5 a
Bacillus spp.	1.8×10^6 a	7.2×10^5 b	1.9×10^6 a	2.7×10^5 a
Ammonifying bacteria	2.1×10^7 a	1.8×10^6 a	9.9×10^5 a	1.5×10^6 a
Filamentous fungi	1.1×10^5 a	1.6×10^5 a	7.5×10^4 a	1.5×10^6 a
Yeasts	2.8×10^3 a	3.7×10^3 a		
	mg C-CO_2/kg dry soil		mg C-CO_2/kg dry soil	
Microbial activity	6.5a	6.8a	3.9a	3.0a

Values are the means of three samples. For each season different letters in each row represent significant differences between the sites ($p<0.05$)
[a]Colony forming units

in their abundance between sites neither in their microbial activity (Table 2).

In the spring of the second year, phosphate solubilizing bacteria and ammonifying bacteria were significantly lower at site B than the control (site A). For the other microbial populations, no significant differences were found in their abundance between sites nor in the microbial activity. In the second autumn, no significant differences were observed between sites for any of the variables (Table 3).

Table 3 Microbiological variables at sites A and B before and after summer irrigation (Year 2)

Variable	Year 2			
	Spring (before summer irrigation)		Autumn (after summer irrigation)	
	Site A (control plot)	Site B (vinasse 150 m^3/ha)	Site A (control plot)	Site B (vinasse 150 m^3/ha)
	CFU[a]/g dry soil		CFU[a]/g dry soil	
Heterotrophic bacteria	1.1×10^7 a	4.3×10^6 a	1.3×10^7 a	3.2×10^7 a
Actinobacteria	3.5×10^6 a	1.7×10^6 a	1.5×10^6 a	1.8×10^6 a
N fixing bacteria	4.6×10^5 a	3.9×10^5 a	1.4×10^4 a	1.2×10^4 a
Pseudomonas fluorescent	8.2×10^3 a	8.1×10^3 a	2.2×10^5 a	2.7×10^5 a
Phosphate solubilizing bacteria	1.4×10^5 a	4.3×10^4 b	4.5×10^4 a	3.7×10^4 a
Bacillus spp.	1.9×10^6 a	7.9×10^5 a	1.6×10^6 a	1.6×10^6 a
Ammonifying bacteria	1.1×10^7 a	9.2×10^5 b	5.1×10^6 a	2.0×10^6 a
Filamentous fungi	1.1×10^5 a	1.0×10^5 a	6.8×10^4 a	2.2×10^5 a
Yeasts	3.7×10^4 a	2.2×10^4 a	1.2×10^4 a	6.8×10^4 a
	mg C-CO_2/kg dry soil		mg C-CO_2/kg dry soil	
Microbial activity	6.1a	4.4a	2.2a	1.0a

Values are the means of three samples. For each season different letters in each row represent significant differences between the sites ($p<0.05$)
[a]Colony forming units

No significant differences were observed in total bacteria abundance between sites in any sampling time (Table 4) when quantified by qPCR.

Controlled assay

In this assay, the dynamics of microbial populations potentially involved in vinasse degradation were observed. The abundance of heterotrophic bacteria and filamentous fungi increased in both soils with experimental vinasse addition respect to the control. In soil from site B (with history of vinasse application in the field) this response was statistically significant only at 14 days for heterotrophic bacteria, and up to 180 days for filamentous fungi (Fig. 2). In soil from site A (without history), the abundance of both populations remained higher with vinasse than the control in mostly samples (Fig. 2). Actinobacteria and ammonifying bacteria populations were not altered with the treatments until 117 days in both soils (Fig. 3). However, at day 180 a significant increase was observed in ammonifying bacteria populations that received a vinasse dose of 125 m^3/ha compared to the control in both soils from sites A and B (Fig. 3).

Discussion

Microbial populations of soils treated with vinasse increased compared to the control under controlled conditions. Heterotrophic bacteria populations in soils from site A, as well as those of filamentous fungi in soils from both sites, remained higher than the control for some time. In soils from site A, this difference was maintained throughout the trial for both populations, while in soils from site B only for the filamentous fungi populations. This difference could be due to the

Table 4 Total bacterial abundance in soil samples from sites A and B

Total bacteria abundance (N° of copies/g dry soil)				
Treatment	Year 1		Year 2	
	Spring	Autumn	Spring	Autumn
Site B (Vinasse 150 m^3/ha)	1.1×10^8 a	1.1×10^8 a	1.6×10^8 a	1.7×10^8 a
Site A (Control plot)	9.9×10^7 a	2.2×10^8 a	1.6×10^8 a	3.8×10^8 a

Values are the means of three samples. For each year and season different letters represent significant differences between the sites ($p<0.05$)

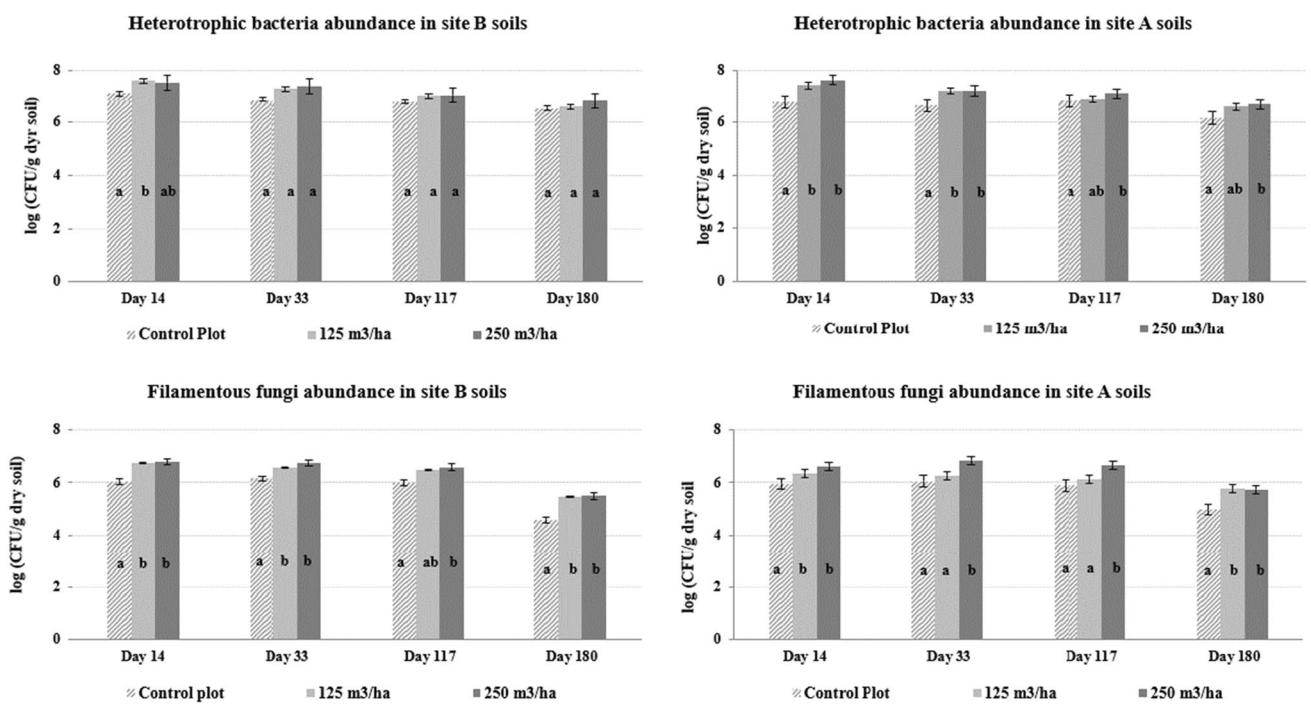

Fig. 2 Heterotrophic bacteria and filamentous fungi counts from soils samples (sites A and B) with different treatments. Values are the means ($n=3$) and error bars indicate standard deviation. For each sampling moment, different letters indicate significant differences compared to control plot ($p<0.05$, Tukey's honestly significant difference, HSD)

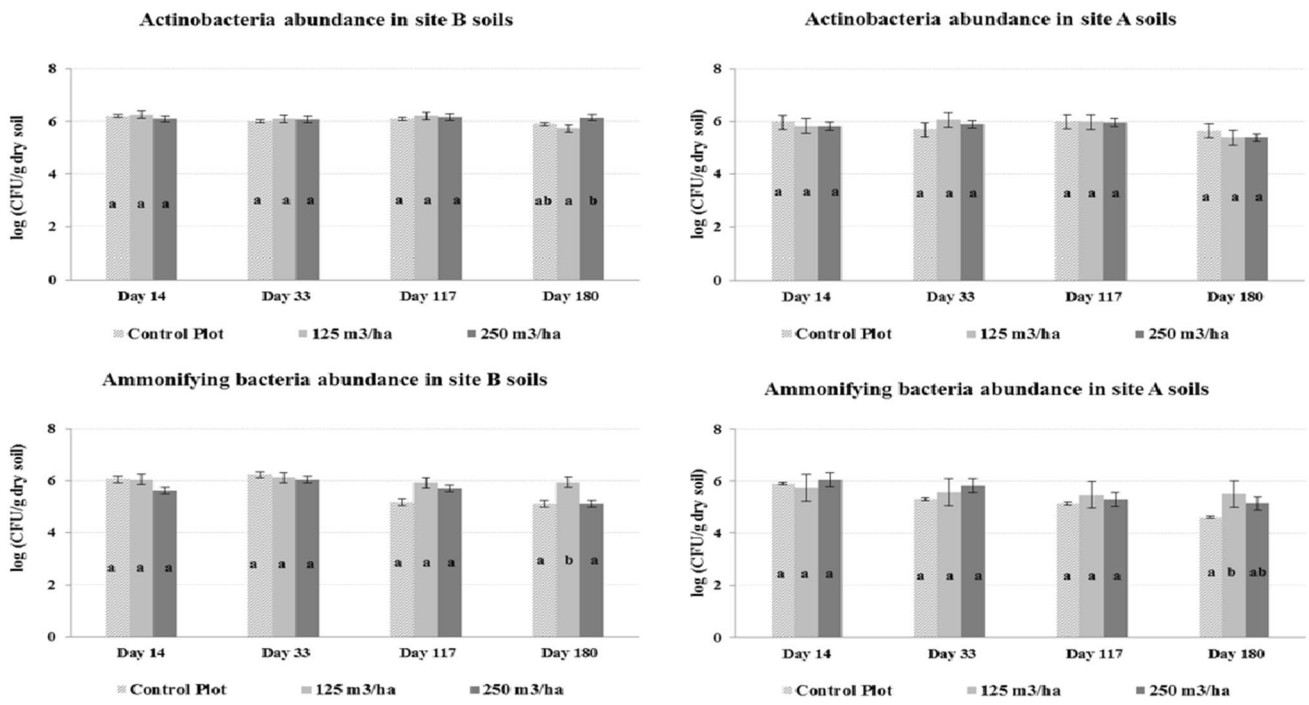

Fig. 3 Actinobacteria and ammonifying bacteria counts from soils samples (sites A and B) with different treatments. Values are the means ($n=3$) and error bars indicate standard deviation. For each sampling moment, different letters indicate significant differences compared to control plot ($p<0.05$, Tukey's honestly significant difference, HSD)

sustained use of vinasse in the field at site B and/or it could be due to a greater tolerance and/or use of the fungi against products present in the vinasse such as phenols (Rulli et al. 2020; España-Gamboa et al. 2017). However, under field conditions, populations of heterotrophic bacteria and filamentous fungi were similar in both sites (vinasse-treated and vinasse-untreated soils). The different results observed between controlled and field conditions could be due to sampling time or the dose of vinasse applied, as well as the variability inherent in this type of test. Crivelaro et al. (2010) also observed that vinasse increases the populations of certain microbial groups under controlled conditions. These results suggest that the microorganisms used vinasse as a C source, thus justifying the application of vinasse as a soil improver. The increase of organic matter in soil can affect several of the critical soil health indicators such as organic matter level, aggregate stability, water infiltration, erosion, and acidification being these parameters fundamental to maintain soil structure and functioning. Changes in these soil properties invariably modify the niches occupied by soil microorganisms, resulting in changes in the content, composition, and activities of the microbial community (Yarwood 2021). The lower the amount of C lost through respiration and the higher the C fraction of the substrate incorporated into the biomass, the more efficient the function of microorganisms as soil improvers (Insam 1990). In turn, if the high microbial activity generated by organic matter application cannot be transformed into biomass due to lack of essential nutrients such as P, N, S, among other micronutrients, the degradation of the soil's own organic matter could occur in a process known as "priming effect" (Bernal et al. 1998a, b).

The study of specific bacterial groups related to organic matter decomposition showed that, under controlled conditions, actinobacteria and ammonifying bacteria populations increased at 117 and 180 days after vinasse application, respectively. Thus, these groups of bacteria may be the mainly responsible for vinasse decomposition. The increase in ammonifying bacteria as a consequence of the application of organic matter residues suggests a transformation of organic to inorganic N, increasing the availability of this nutrient for plants (Kandeler et al. 1994). The increase in actinobacteria population, due to vinasse application, could be considered a positive impact because these soil inhabiting bacteria play important roles in various ecological processes such as organic matter decomposition and toxic pollutant and heavy metal bioremediation, thus contributing to the restoration of soil fertility and environmental sustainability (Shivlata and Satyanarayana 2017).

Regarding the amount of phosphate solubilizing bacteria (PSB), under field conditions a decrease was observed in the treatment with vinasse in spring. Likewise, Widdig et al. (2019) observed that the relative abundance of phosphate solubilizing bacteria (PSB) was significantly lower in soils that received treatments with N and N + P compared

to controls. Therefore, nutrient availability seems to negatively affect this functional trait of the microbial community. Moreover, in beech rhizosphere soil, the efficiency of P solubilization was lower in nutrient-rich soils compared to soils with low nutrient availability (Nicolitch et al. 2016). Therefore, microorganisms capable of solubilizing P may have an advantage in conditions of scarcity of available nutrients, thus facilitating its uptake by plants (Widdig et al. 2019). Although fertilizers or amendments application can promote plant growth, they can also inhibit some bacterial growth.

Another parameter that was determined to estimate the effect of vinasse application on soil microorganisms was microbial activity, which is related to the rate of decomposition of different substrates by soil microorganisms. In the field conditions, no significant differences were observed in respiration between vinasse treated soils compared to the control soil in any of the samplings. However, under controlled conditions in the same experiment we analyzed, Takata (2020) determined a temporary increase in microbial respiration due to vinasse application in soils from both sites up to 20 days, and a decrease over time. This result, together with the increase in microbial populations determined in the same soils and under controlled conditions, indicates that the vinasse decomposes rapidly and is effectively incorporated into the biomass.

This temporary increase in microbial activity can also be observed in the field, depending on the type of soil, as reported by Senatore (2013). In turn, Bustamante et al. (2011) and Prata et al. (2001) also observed an increase in microbial activity in soils that received vinasse. Tripathi (2011), when applying distillery residue sludge to tropical soils for agriculture observed, as with the vinasse application, an increase in respiration between 30 and 60 days and then a gradual decrease in activity over time.

In addition, the entire bacterial population (*Bacteria* domain) existing in both sites was quantified by analyzing the gene that codes for the rRNA of the minor ribosomal subunit (16S rRNA). The abundance of total bacteria found did not show significant differences between vinasse-treated and vinasse-untreated soils at any sampling time. These results are similar to those obtained in Uruguay in other types of soils destined for sugarcane cultivation, in which the total abundance of bacteria was evaluated in response to the application of vinasse (Senatore 2013). Likewise, assays carried out on other crops with other amendments and/or fertilizers did not show significant differences in total bacteria population, such as in potato crops (Dandie et al. 2008).

The bacterial groups that registered significant differences with respect to their abundance in some sampling times, would represent a small fraction of the total bacterial populations present in the soils, since their quantification by Real Time PCR did not show any significant changes throughout the test. Taking together the findings from us and others (Chavarria et al. 2018; Dandie et al. 2008; He et al. 2007; Okano et al. 2004), the total bacteria population is generally not affected by the application of fertilizers. This could be either because the changes in specific populations are small relative to the total population to be detected by the entire bacterial populations, or because it is difficult to detect in a large scale of time.

Conclusions

Due to the great complexity of soil microbiota, multiple methods are necessary to study its changes, since different methodologies can give us information about the occurrence of different phenomena.

This became evident in our study, where the results obtained regarding total bacteria determination indicated that vinasse does not have a significant incidence on the soil microbiota as a whole, while it did have a significant impact on microbial activity and on some groups of microorganisms such as actinobacteria and ammonifying bacteria that represent part of the microbiota capable of using and degrading the vinasse nutrients.

We understand that the application of vinasse in terms of its use as a soil improver can have different effects depending on soil type and conditions in which it is applied. The effect can be positive if soil microbiota increases due to the contribution of nutrients from the vinasse, or negative if population decreases or its growth is restricted by other nutrients or the physical–chemical condition of the soil and only respiration increases.

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Author contributions Conception or design: DS and NB; performed the experiments and acquisition of data: DS and AQ; analysis and interpretation of data: DS, AQ, NB and JM; drafting of the manuscript: DS, NB and JM; statistical analysis: DS and AQ; obtaining funding: NB; supervising the work: NB and JM; coordinating the research project: NB, wrote the paper: DS, NB and JM.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval This research not involve human participants and/or animals.

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